SCIENCE DISCOVERY LAB

Basic 1 — Learner’s Guide

29 curriculum-aligned lessons • Observe • Explore • Explain • Practise

A colourful first science book about the natural world, materials, weather, water, energy, our bodies and safe everyday living.

Use this book with a teacher, parent or another trusted adult. All written explanations and questions are original educational material based on the uploaded NaCCA Basic 1 Science curriculum.

Your discovery journey

Diversity of Matter

1. Look Closely at Our Surroundings • 2. Plants and Animals in Ghana • 3. What Living Things Need • 4. Living, Non-living and Once-living • 5. Materials Around Us • 6. Sorting Materials by Their Features • 7. Solids, Liquids and Gases • 8. Mixing and Separating

Cycles

9. Day and Night Repeat • 10. The Sun Gives Us Light • 11. Where Did the Puddle Go? • 12. Water Sources and Uses • 13. Air Is All Around Us

Systems

14. My Body Parts • 15. Homes of Living Things

Forces and Energy

16. Energy Helps Us Do Things • 17. Hot, Warm and Cold • 18. Electricity Around the Home • 19. Electronic Devices and Their Jobs • 20. Pushes and Pulls • 21. Simple Machines Help Us

Humans and the Environment

22. Bathing to Stay Clean • 23. Keep Teeth Clean • 24. Wash Hands the Right Way • 25. Keep Our Environment Clean • 26. Care for Healthy Skin • 27. Technology Helps Communities • 28. Keeping Food for Later • 29. Weather Watchers

Safety comes first

Do not touch broken glass, unknown substances, electrical sockets or hot objects. Never taste materials during experiments. Water, food and hygiene activities need appropriate adult guidance. During storms, stay in safe shelter.

QUEST 1: Look Closely at Our Surroundings

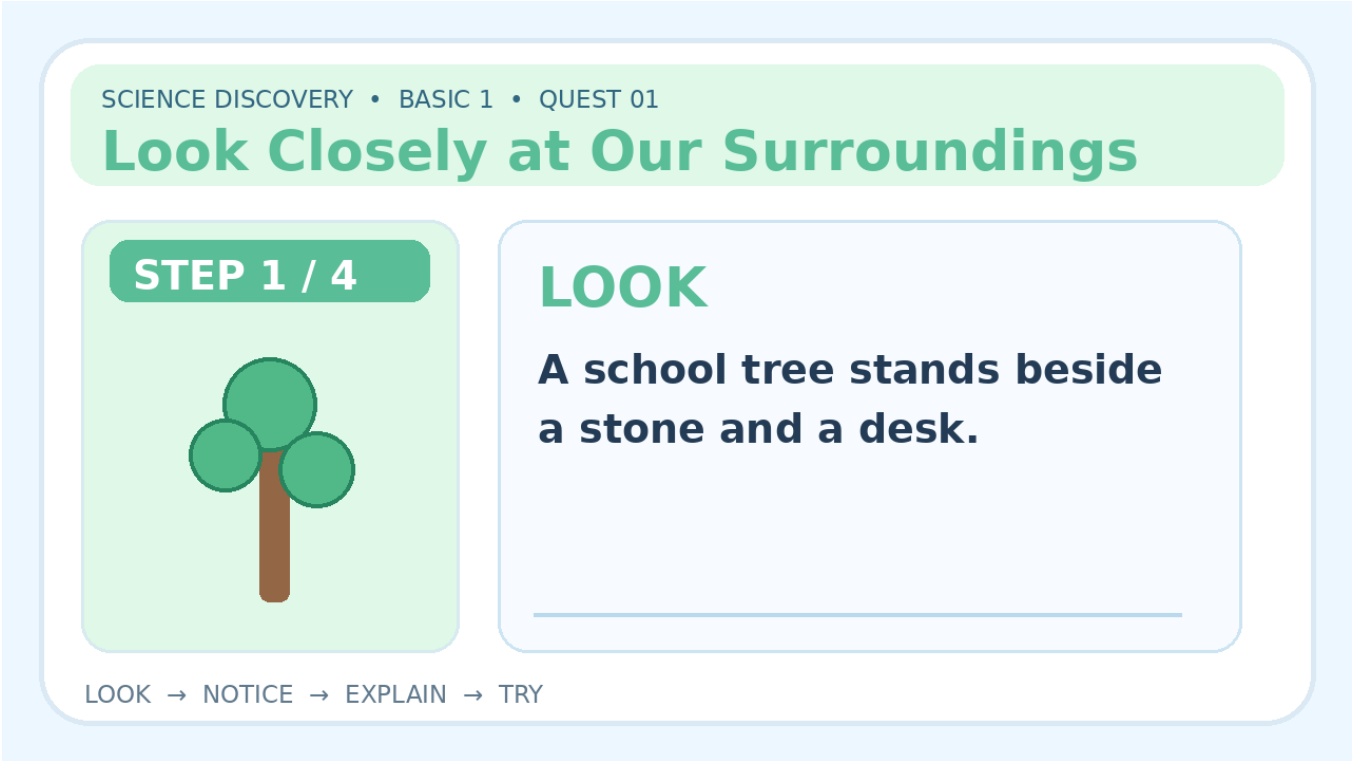
Diversity of Matter | Observe and describe different things around you.

LET’S LEARN

Our school and community have many different things. A tree has leaves and a stem. A stone feels hard. We use our eyes to notice colours and shapes, and we can describe what we see. Scientists observe before grouping things. Look carefully without disturbing animals or touching unknown objects.

Science words to remember

Observe carefully • Describe colour and shape • Sort safely



Try it for yourself

Take a supervised walk in the school compound. Draw two living things and two non-living things. Tell a friend one detail about each.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 1: Show what you know

Observe the four teaching steps

1. LOOK: A school tree stands beside a stone and a desk.

2. OBSERVE: See green leaves, a grey stone and a brown desk.

3. SORT: Put tree in the living group; stone and desk in non-living.

4. SHOW: Describe one colour, shape or texture for each item.

Guided practice — answer in your exercise book

1. What do you use to see a red flower?
2. What is one observable feature of a leaf?
3. Which item can you find growing in a garden?
4. Which of these may be rough to touch?
5. Which item belongs to a group of living things?
6. Which item is usually a non-living object?
7. What should you do before drawing a school tree?
8. What is the safest way to observe an unknown insect?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 2: Plants and Animals in Ghana

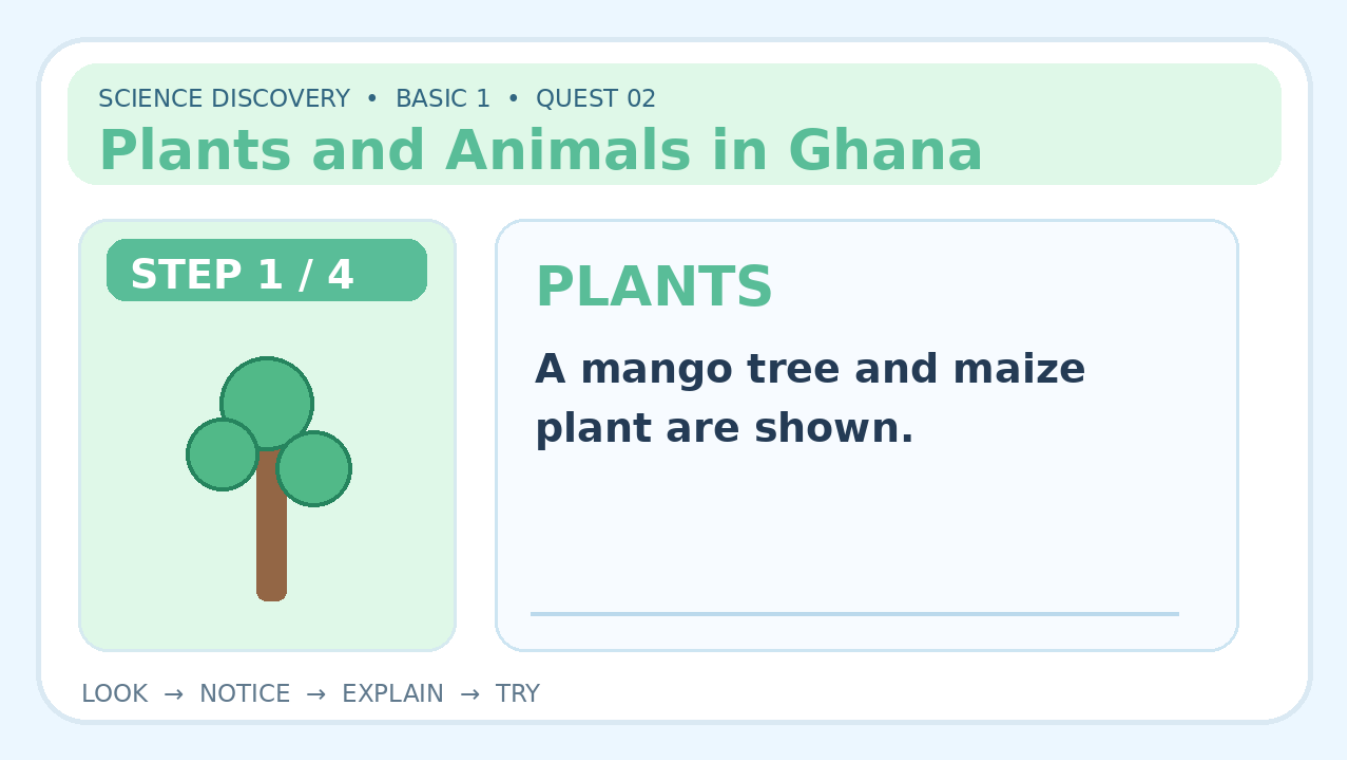
Diversity of Matter | Name common plants and animals in your community.

LET’S LEARN

A maize plant and a mango tree are plants. Goats, chickens, dogs and butterflies are animals. We may see plants on farms, in gardens or beside a house. Animals live in different places and move in different ways. We can learn English and local names for familiar living things.

Science words to remember

Identify plant • Identify animal • Compare local examples



Try it for yourself

With an adult, name three local plants and three animals. Draw one plant and one animal and label each.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 2: Show what you know

Observe the four teaching steps

1. PLANTS: A mango tree and maize plant are shown.

2. ANIMALS: A chicken and a goat appear beside them.

3. COMPARE: Plants have stems and leaves; many animals move on legs.

4. TRY: Name and draw one Ghanaian plant and animal.

Guided practice — answer in your exercise book

1. Which is a plant?
2. Which is an animal?
3. Which grows as a tree?
4. Which can lay eggs?
5. Which is a farm crop?
6. Which is often kept as livestock?
7. Which living thing may have a beak?
8. Which is a plant part seen on a mango tree?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 3: What Living Things Need

Diversity of Matter | Identify food, water, air and other basic needs of living things.

LET’S LEARN

Plants and animals need suitable conditions to stay alive. Animals need water, air and food. Plants need water, air, light and nutrients to grow. Living things also need an appropriate place to live. We should give pets clean water and protect seedlings without overwatering them.

Science words to remember

Needs of animals • Needs of plants • Care for living things

SCIENCE DISCOVERY • BASIC 1 • QUEST 03

What Living Things Need

STEP 1 / 4



MEET

A goat, a child and a small green plant.

LOOK → NOTICE → EXPLAIN → TRY

Try it for yourself

Water a small plant with an adult. Make a picture chart showing its needs: water, air and sunlight.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 3: Show what you know

Observe the four teaching steps

1. MEET: A goat, a child and a small green plant.

2. ADD: Place water and air symbols near all living things.

3. GROW: Show the plant getting sunlight and sensible watering.

4. CARE: Match each living thing to a suitable need.

Guided practice — answer in your exercise book

1. What must a goat drink?
2. What do people breathe?
3. Which is food for a goat?
4. What helps a green plant grow?
5. What should we give a thirsty pet?
6. What should a seedling receive?
7. What can protect an animal during heavy rain?
8. Which is needed by both plants and animals?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 4: Living, Non-living and Once-living

Diversity of Matter | Explain how living things differ from non-living objects.

LET’S LEARN

Living things grow and carry out life processes. A kitten eats and grows. A chair does not eat or grow. A dried fallen leaf came from a living tree, but the fallen leaf is no longer alive. A moving toy car may move when pushed, yet it is not alive. Movement alone does not make an object living.

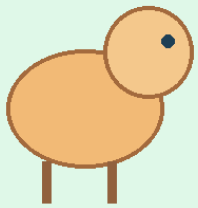
Science words to remember

Living • Never alive • Once alive

SCIENCE DISCOVERY • BASIC 1 • QUEST 04

Living, Non-living and Once-living

STEP 1 / 4



WATCH

A kitten eats and grows beside a toy car.

LOOK → NOTICE → EXPLAIN → TRY

Try it for yourself

Create three circles: living, never alive, once living. Sort drawings of a cat, plastic spoon and dry leaf.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 4: Show what you know

Observe the four teaching steps

1. WATCH: A kitten eats and grows beside a toy car.

2. CHECK: The car moves when pushed but does not feed or grow.

3. COMPARE: A fresh growing plant and a dry fallen leaf.

4. SORT: Living, once-living and never-living objects.

Guided practice — answer in your exercise book

1. Which is alive now?
2. Which is non-living and never alive?
3. Which was once part of something living?
4. Why is a toy car not a living thing?
5. Which item needs food to grow?
6. Which can make more of its kind as it grows older?
7. Which was never alive?
8. Which needs air for life?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 5: Materials Around Us

Diversity of Matter | Identify materials used to make everyday objects.

LET’S LEARN

Things are made from materials. A desk can be made of wood or metal. A window may be made of glass. A water bottle might be plastic or glass. Clothes are made from fabric. Materials have properties that make them useful. Observe objects carefully and do not touch sharp glass.

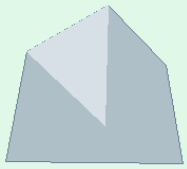
Science words to remember

Identify materials • Match objects • Safe handling

SCIENCE DISCOVERY • BASIC 1 • QUEST 05

Materials Around Us

STEP 1 / 4



SHOW

Wood, metal, fabric and plastic appear as labelled materials.

LOOK → NOTICE → EXPLAIN → TRY

Try it for yourself

Find one safe item each made mainly from wood, plastic, metal and fabric. Draw them.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 5: Show what you know

Observe the four teaching steps

1. SHOW: Wood, metal, fabric and plastic appear as labelled materials.

2. MATCH: A wooden desk is linked to wood.

3. MATCH: A metal spoon is linked to metal.

4. TRY: Identify the main material of an object near you.

Guided practice — answer in your exercise book

1. A wooden chair is made mainly of what?
2. Which material is common in spoons?
3. A clear drinking windowpane may be made of what?
4. What material is in a plastic bowl?
5. A cotton shirt is an example of what material?
6. Which material is used to make many pots?
7. What is safer than holding broken glass?
8. Which object may be made of clay?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 6: Sorting Materials by Their Features

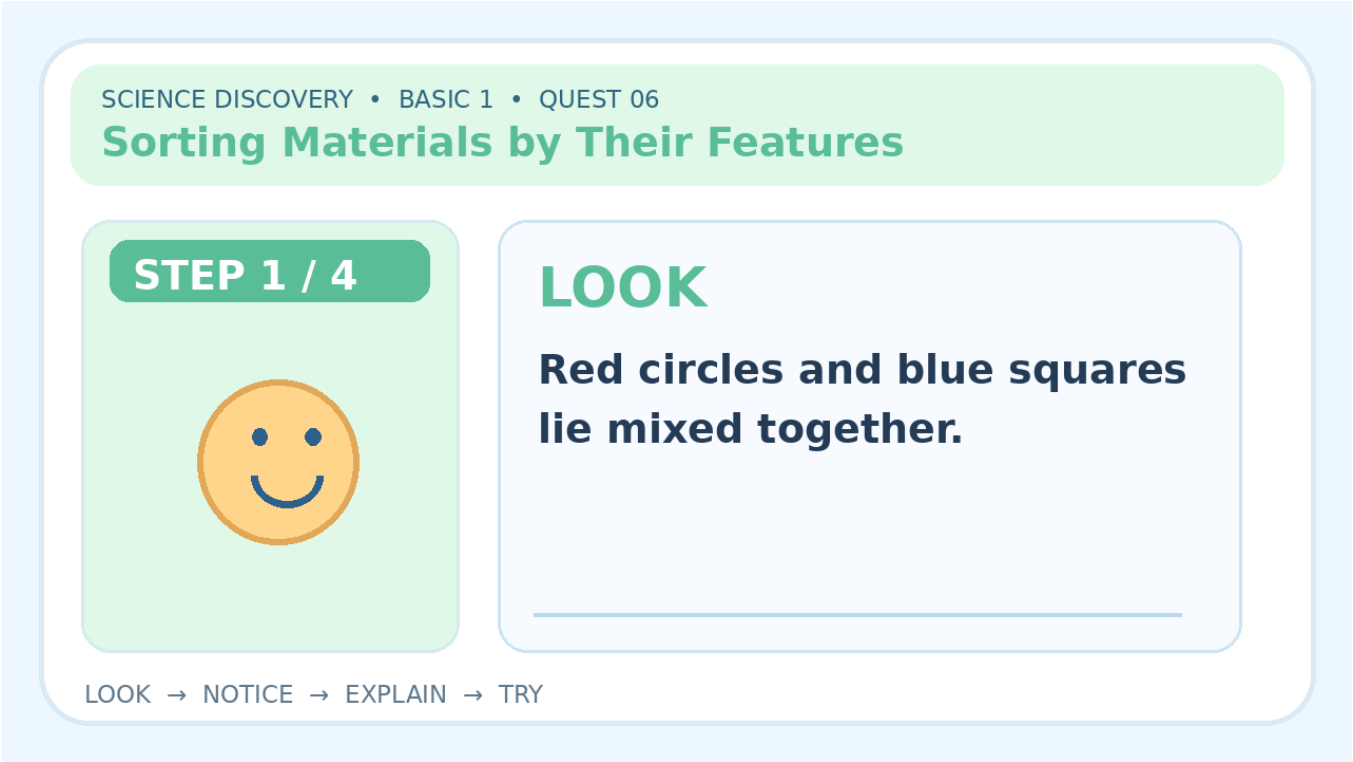
Diversity of Matter | Group materials by colour, shape, size and texture.

LET’S LEARN

We can describe materials by observable properties. Some are smooth and some are rough. Some are hard, some are soft; some are big, others small. A red stone and a red cup share a colour but are not made from the same material. Use one clear sorting rule at a time.

Science words to remember

Colour • Texture • Size and shape



Try it for yourself

Use safe bottle tops or paper shapes to sort by colour. Sort them again by size. Explain why the groups changed.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 6: Show what you know

Observe the four teaching steps

1. LOOK: Red circles and blue squares lie mixed together.

2. SORT COLOUR: Put red pieces in one group and blue in another.

3. SORT SHAPE: Make a new circle group and square group.

4. EXPLAIN: The same items can be grouped using different features.

Guided practice — answer in your exercise book

1. Which words describe texture?
2. How can you group a red cup and red ball?
3. Which is usually soft?
4. Which is usually rough?
5. What rule sorts large and small stones?
6. What rule sorts circles and squares?
7. Which material is often hard?
8. Why might two groups change after sorting again?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 7: Solids, Liquids and Gases

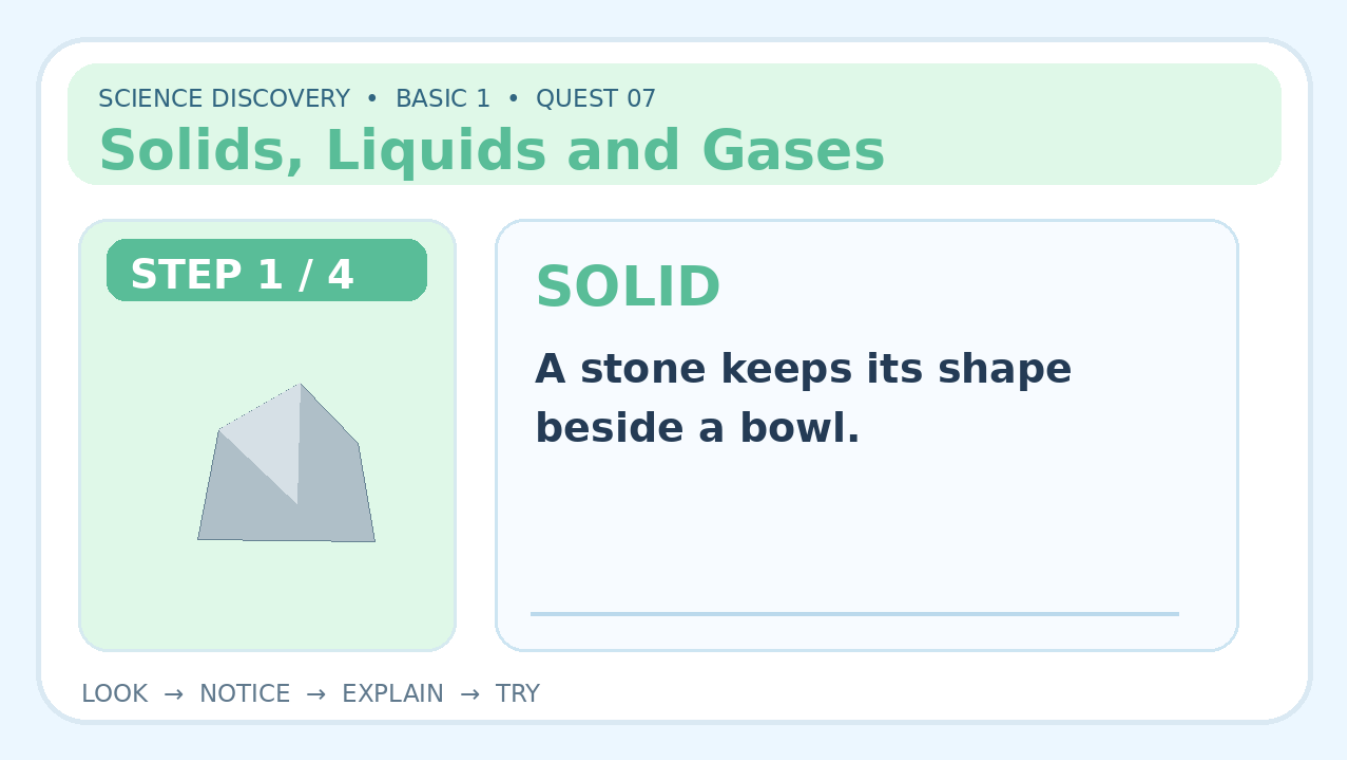
Diversity of Matter | Sort common materials into solids, liquids and gases.

LET’S LEARN

A solid usually keeps its shape, such as a stone or spoon. A liquid can be poured and takes the shape of its container, such as drinking water. Air is a gas: it spreads out and is often invisible. We must not taste unknown liquids or breathe gases on purpose.

Science words to remember

Solids • Liquids • Gases



Try it for yourself

With adult supervision, observe a stone, a cup of water and a blown-up balloon. Draw each and label its state.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 7: Show what you know

Observe the four teaching steps

1. SOLID: A stone keeps its shape beside a bowl.

2. LIQUID: Water flows from a jug into a cup.

3. GAS: Air fills a balloon and makes it larger.

4. SORT: Stone = solid; water = liquid; air = gas.

Guided practice — answer in your exercise book

1. What state is a stone?
2. What state is clean water in a cup?
3. What state is the air in a balloon?
4. Which can flow into a different shaped container?
5. Which usually keeps its own shape?
6. Which substance is often invisible but fills space?
7. Which can you pour from a bottle?
8. What should you do with an unknown liquid?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 8: Mixing and Separating

Diversity of Matter | Understand that a mixture contains two or more materials.

LET’S LEARN

A mixture has two or more substances or objects together. Sand mixed with small stones is one example. Rice mixed with beans is another. We can often separate visible solids by hand or with a sieve when it is safe to do so. Never mix cleaning chemicals or unknown liquids.

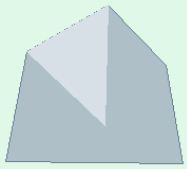
Science words to remember

Identify mixtures • Make mixtures • Separate safely

SCIENCE DISCOVERY • BASIC 1 • QUEST 08

Mixing and Separating

STEP 1 / 4



BEFORE

Red paper circles and blue squares are in different bowls.

LOOK → NOTICE → EXPLAIN → TRY

Try it for yourself

Mix clean paper circles and squares. Separate them by shape. Describe what changed and what stayed the same.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 8: Show what you know

Observe the four teaching steps

1. BEFORE: Red paper circles and blue squares are in different bowls.

2. MIX: Pour the paper shapes into one tray.

3. SEPARATE: Place circles and squares back into two groups.

4. TELL: The materials did not change into something else.

Guided practice — answer in your exercise book

1. Which shows a mixture?
2. A bowl of sand and pebbles contains how many types of material?
3. What can separate large pebbles from fine sand?
4. Which is safer for a classroom mixing activity?
5. What is one way to separate beans from large stones?
6. Which two items can make a safe paper mixture?
7. What happens when you mix two kinds of beads?
8. Which item is not a mixture of two things?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 9: Day and Night Repeat

Cycles | Describe the repeating cycle of day and night.

LET’S LEARN

The sun appears in our sky during the day. At night the sky is often darker. Day and night come again and again. We do different activities at different times; some people work at night. A daily routine might include going to school in the day and sleeping at night.

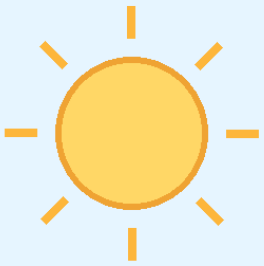
Science words to remember

Day • Night • Repeating cycles

SCIENCE DISCOVERY • BASIC 1 • QUEST 09

Day and Night Repeat

STEP 1 / 4



SEE

A bright sun shines over a school.

LOOK → NOTICE → EXPLAIN → TRY

Try it for yourself

Draw a simple two-picture diary of something you do in daylight and something you do after sunset.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 9: Show what you know

Observe the four teaching steps

1. SEE: A bright sun shines over a school.

2. CHANGE: The sky darkens after sunset.

3. REPEAT: Night is followed by another day.

4. ORDER: Day → night → day → night.

Guided practice — answer in your exercise book

1. When is the sky usually bright from sunlight?
2. What often follows daytime?
3. What comes after night in the repeating pattern?
4. Which is a common day activity for a learner?
5. Which object may be seen in the day sky?
6. Which part of the cycle is usually dark?
7. How often do day and night happen?
8. Which pattern is correct?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 10: The Sun Gives Us Light

Cycles | Identify the Sun as the main natural light source for Earth.

LET’S LEARN

The Sun gives Earth natural light during the day. Daylight helps us see outdoors. A torch or lamp can provide light when it is dark. A source of light makes light; a mirror only reflects light. Never look directly at the Sun.

Science words to remember

Natural light • Artificial light • Safe observation



Try it for yourself

Compare a bright place in the shade with a dark indoor corner. Tell an adult which receives more daylight.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 10: Show what you know

Observe the four teaching steps

1. SOURCE: The Sun shines over a classroom.

2. SEE: A learner reads in daylight near a window.

3. COMPARE: A torch lights up a dark room.

4. SAFETY: Observe the daylight; never stare at the Sun.

Guided practice — answer in your exercise book

1. What is the main natural source of daylight?
2. Which can give light during a power cut?
3. What should you avoid?
4. Which is an artificial light source?
5. What does a mirror do with light?
6. When does sunlight usually brighten the school compound?
7. Which source gives light without a battery?
8. Why can you see a book better near a bright window?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 11: Where Did the Puddle Go?

Cycles | Observe how rainwater puddles can disappear.

LET’S LEARN

After rain, water may collect in puddles. Later, sunlight, warmth and moving air can help some liquid water change into water vapour in the air. This is called evaporation. Some puddles also soak into the ground or flow away. We should observe, not claim that all puddles disappear for only one reason.

Science words to remember

Puddles • Evaporation • Other causes



Try it for yourself

With an adult, mark the edge of a safe puddle after rain. Observe it from a distance later. Record whether it became smaller.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 11: Show what you know

Observe the four teaching steps

1. AFTER RAIN: A shallow puddle lies on a sunny path.

2. LATER: The outline of the puddle is smaller.

3. EXPLAIN: Some water evaporates into invisible vapour.

4. ALSO: Some water may soak into soil or flow away.

Guided practice — answer in your exercise book

1. What may form on the ground after rain?
2. What can happen to a puddle on a warm day?
3. What is evaporation?
4. Which condition can help water evaporate?
5. Where can some rainwater go besides evaporating?
6. What can move puddle water away?
7. What should we do before explaining a puddle change?
8. What is a safe way to study a puddle?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 12: Water Sources and Uses

Cycles | Identify where we get water and how we use it.

LET’S LEARN

Water comes from rain, rivers, lakes, wells, boreholes and treated piped supplies. We use suitable safe water for drinking, cooking, washing and cleaning. Not every water source is safe to drink without treatment. We should turn taps off and avoid wasting water.

Science words to remember

Sources • Uses • Water safety



Try it for yourself

Draw a tap, a borehole or another local water source. Ask an adult how drinking water is kept safe at home.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 12: Show what you know

Observe the four teaching steps

1. FIND: Rain, a borehole and a tap are shown as sources.

2. USE: A child washes hands using safe water.

3. SAVE: A tap is turned off after use.

4. CHECK: An adult makes sure drinking water is safe.

Guided practice — answer in your exercise book

1. Which is a water source?
2. Which is a common use of water?
3. Which source falls from clouds?
4. What should you do with tap water when finished?
5. Which water should be used for drinking?
6. Which can supply water to some communities?
7. Which activity needs water?
8. What should we do before drinking water from an uncertain source?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 13: Air Is All Around Us

Cycles | Show that air exists even when we cannot see it.

LET’S LEARN

Air is around us even though we usually cannot see it. A balloon fills with air when blown up. Moving air can make flags and leaves flutter. We breathe air to live. Use clean air and never cover anyone’s face with a plastic bag.

Science words to remember

Air occupies space • Moving air • Breathing



Try it for yourself

With adult help, blow up a clean balloon or wave a sheet of paper to feel moving air. Compare before and after.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 13: Show what you know

Observe the four teaching steps

1. BEFORE: An empty balloon is small.

2. INFLATE: Air enters and makes the balloon larger.

3. MOVE: A paper flag bends when air blows.

4. EXPLAIN: Invisible air takes up space and can move things.

Guided practice — answer in your exercise book

1. What fills a blown-up balloon?
2. How can you feel moving air?
3. What may flutter when wind blows?
4. What do humans need to breathe?
5. Why does a balloon become bigger when inflated?
6. Which is a safe air activity?
7. Can we usually see clean air itself?
8. What can move leaves on a calm tree suddenly?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 14: My Body Parts

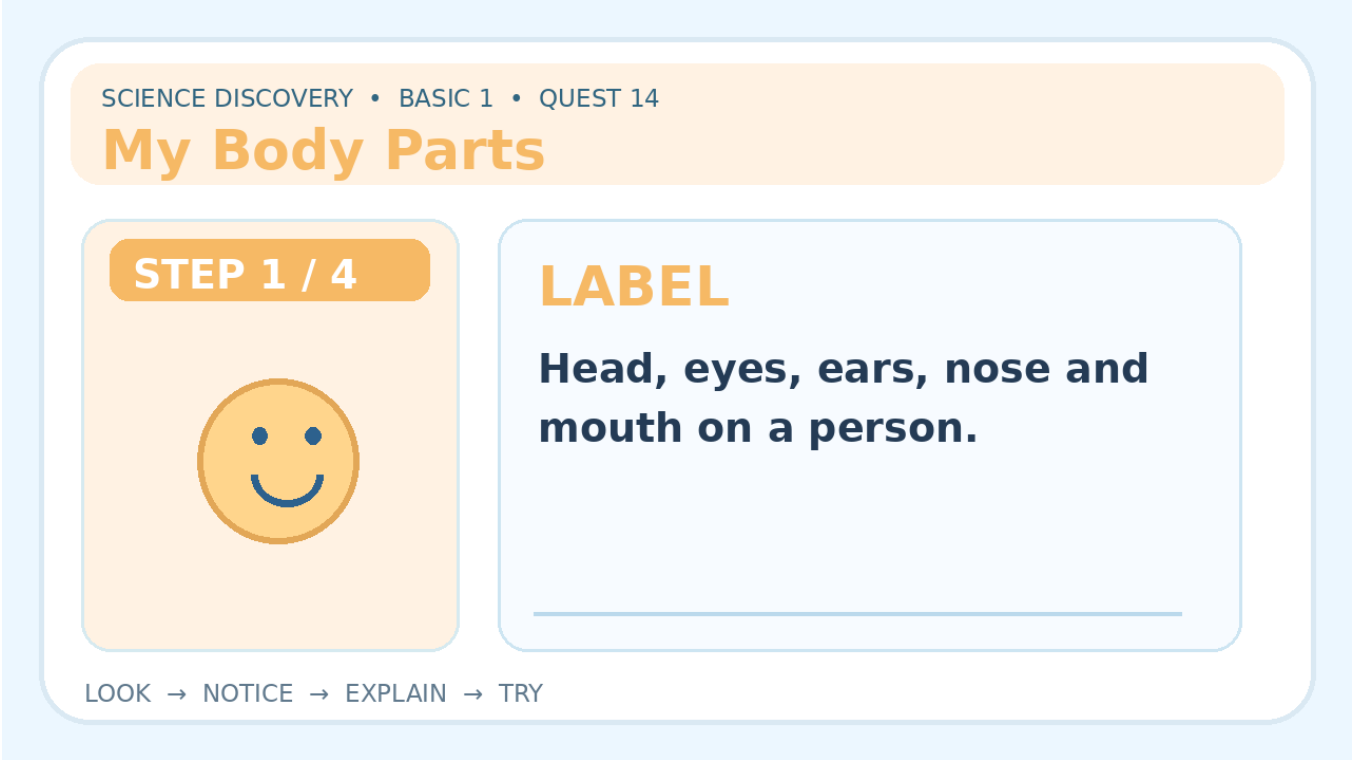
Systems | Name visible parts of the human body and their uses.

LET’S LEARN

Our bodies have different visible parts. We see with our eyes, hear with our ears and smell with our nose. Hands can grasp objects, and legs help us walk. We care for all parts and respect that bodies can look and move differently.

Science words to remember

Names • Sense organs • Movement



Try it for yourself

Draw a person and label head, eyes, ears, nose, hands and legs. Point to your own body parts as you name them.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 14: Show what you know

Observe the four teaching steps

1. LABEL: Head, eyes, ears, nose and mouth on a person.

2. DO: Eyes look at a picture and ears listen to a bell.

3. MOVE: Hands clap; legs walk.

4. CARE: Every body deserves respect and kindness.

Guided practice — answer in your exercise book

1. Which part helps you see?
2. Which part helps you hear?
3. Which part helps you smell?
4. Which parts help you clap?
5. Which body parts help most people walk?
6. Where are your eyes?
7. Which part can hold a pencil?
8. How should we treat people with different bodies?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 15: Homes of Living Things

Systems | Match organisms to suitable places where they live.

LET’S LEARN

The place where an organism lives is its habitat. Fish live in water. Goats often live on land. Birds may use nests in trees and fly through air. Habitats provide water, food and shelter. We should avoid destroying animal homes.

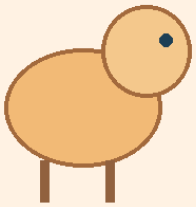
Science words to remember

Land • Water • Trees and air

SCIENCE DISCOVERY • BASIC 1 • QUEST 15

Homes of Living Things

STEP 1 / 4



LAND

A goat grazes on grass.

LOOK → NOTICE → EXPLAIN → TRY

Try it for yourself

Observe a safe place near school. Draw one organism and the place where it lives.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 15: Show what you know

Observe the four teaching steps

1. LAND: A goat grazes on grass.

2. WATER: A fish swims in a pond.

3. TREE: A bird rests in a nest.

4. MATCH: Each animal is linked to its suitable home.

Guided practice — answer in your exercise book

1. Where does a fish usually live?
2. Where can a goat walk and graze?
3. Where might a bird build a nest?
4. What is a habitat?
5. What can a habitat provide?
6. Where might a crab live?
7. What should we do to a bird's nest?
8. Where does an earthworm often live?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 16: Energy Helps Us Do Things

Forces and Energy | Recognise uses of energy in daily life.

LET’S LEARN

We need energy to move and work. Our bodies get energy from food. A torch uses energy from batteries to give light. A car uses energy from fuel. We can save energy by switching off electrical devices when we are finished with them.

Science words to remember

Body energy • Light energy • Saving energy



Try it for yourself

Name three activities that need energy. Draw a child walking, a torch shining and a bicycle moving.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 16: Show what you know

Observe the four teaching steps

1. EAT: A child eats a balanced meal.

2. MOVE: The child claps and walks using energy.

3. LIGHT: A torch changes battery energy into light.

4. SAVE: An unused lamp is switched off.

Guided practice — answer in your exercise book

1. What gives our bodies energy to move?
2. What helps a battery torch shine?
3. What does a moving bicycle need?
4. How can we save electricity?
5. Why does a car need fuel?
6. What helps a child run?
7. What can electrical energy do in a fan?
8. Which activity needs energy?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 17: Hot, Warm and Cold

Forces and Energy | Use the words hot, warm and cold correctly.

LET’S LEARN

Things may feel hot, warm or cold. Ice is cold. Tea made with boiling water can be very hot and may burn. Warm water is between hot and cold. We must never touch a pot, flame or unknown hot liquid to test its temperature. Ask an adult to help with safe comparisons.

Science words to remember

Cold • Warm • Hot and safety

SCIENCE DISCOVERY • BASIC 1 • QUEST 17

Hot, Warm and Cold

STEP 1 / 4



COLD

An ice block is shown in a bowl.

LOOK → NOTICE → EXPLAIN → TRY

Try it for yourself

Sort picture cards of ice, warm bathwater and steaming tea. Do not touch hot objects during the activity.

Safety note: Do not touch hot objects, stoves, flames, plugged-in devices or boiling liquids.

QUEST 17: Show what you know

Observe the four teaching steps

1. COLD: An ice block is shown in a bowl.

2. WARM: A bowl of safely warm water is shown.

3. HOT: Steam rises from tea; a stop sign warns not to touch.

4. SORT: Cold, warm and hot picture cards.

Guided practice — answer in your exercise book

1. How does an ice block usually feel?
2. How might newly boiled tea feel?
3. What describes water between hot and cold?
4. What should you do with a steaming kettle?
5. Where can water become cold?
6. What might sunshine do to a stone?
7. How should a child check an unknown hot object?
8. Which word best describes ice cream taken from a freezer?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 18: Electricity Around the Home

Forces and Energy | Identify appliances that need electricity.

LET’S LEARN

Electricity provides energy for some devices. A television, refrigerator and electric fan need an electrical supply. A broom and an ordinary spoon work without electricity. Never touch wall sockets, exposed wires or plugged-in electrical appliances without adult guidance.

Science words to remember

Electrical appliances • Other tools • Electrical safety



Try it for yourself

Make a two-column picture chart: needs electricity / works without electricity. Use familiar home items.

Safety note: Never handle mains plugs, sockets or exposed wires during a science activity.

QUEST 18: Show what you know

Observe the four teaching steps

1. SORT: Television and fan are under Needs electricity.

2. SORT: Broom and spoon are under Does not need electricity.

3. USE: A fan turns to move air when safely powered.

4. SAFETY: Keep hands and metal objects away from sockets.

Guided practice — answer in your exercise book

1. Which needs electricity to run?
2. Which can work without electricity?
3. Which uses electricity to keep food cold?
4. Which can show programmes using electricity?
5. What should you do with an exposed wire?
6. What should you never push into a socket?
7. Which item can give light when powered?
8. Why can electricity be useful?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 19: Electronic Devices and Their Jobs

Forces and Energy | Name simple electronic devices and their functions.

LET’S LEARN

Electronic devices can help us share information, listen and learn. A radio plays news and music. A phone can help people communicate. A calculator helps with calculations. A thermometer measures temperature. We use devices carefully, with appropriate adult guidance.

Science words to remember

Communication • Information • Measurement



Try it for yourself

Look at drawings of a radio, phone, calculator and thermometer. Match each to its main use.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 19: Show what you know

Observe the four teaching steps

1. SEE: A radio shows sound waves for a news report.

2. USE: A phone connects two people.

3. MEASURE: A thermometer shows temperature.

4. MATCH: Choose each device for its appropriate task.

Guided practice — answer in your exercise book

1. What can a radio play?
2. What can a phone help us do?
3. What can a calculator help with?
4. What can a thermometer measure?
5. Which device may help us hear announcements?
6. Which should be used with adult guidance?
7. What might show a video lesson?
8. Which device could help call a family member?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 20: Pushes and Pulls

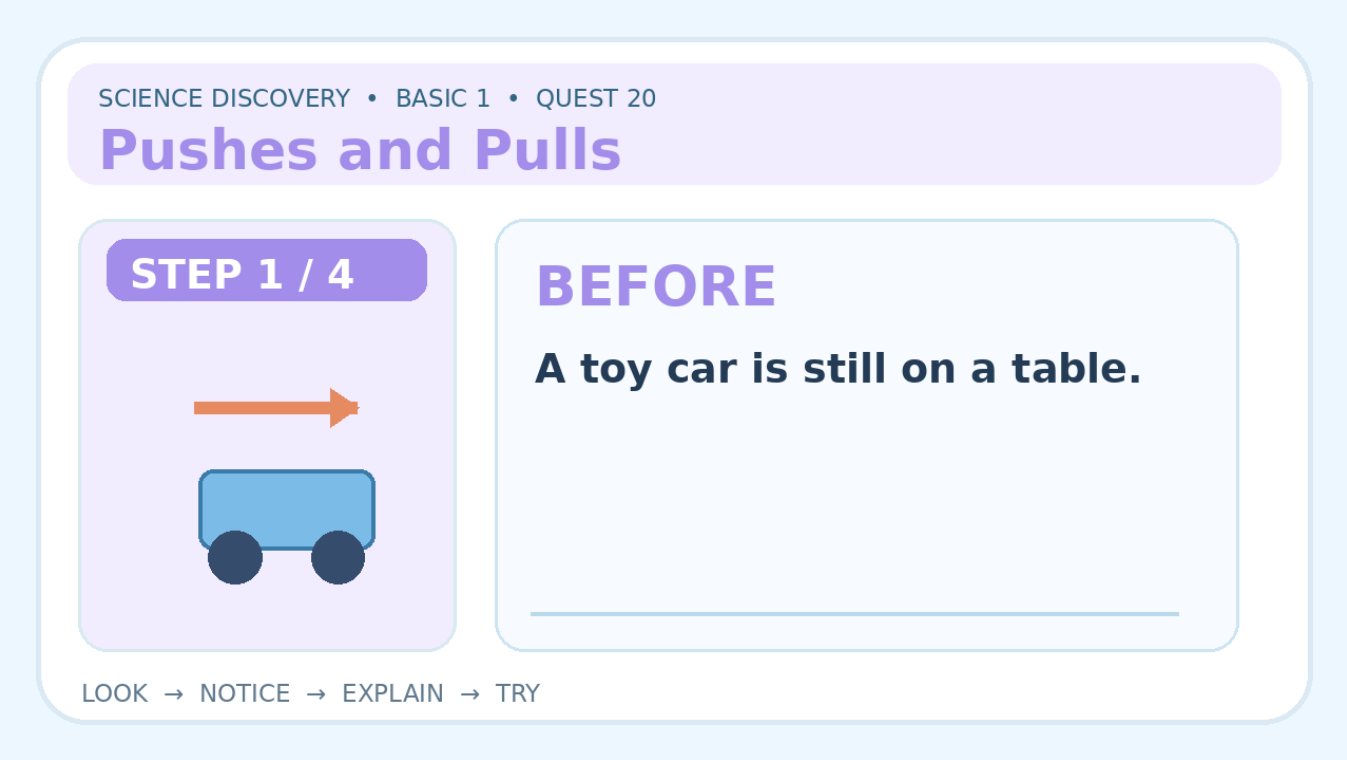
Forces and Energy | Identify a push and a pull as kinds of force.

LET’S LEARN

A force is a push or a pull. Pushing can move a toy car away from you. Pulling can bring a drawer towards you. A force can start, stop or change the movement of an object. Move only light, safe objects when practising.

Science words to remember

Push • Pull • Effects of force



Try it for yourself

Using a light toy car, push it gently along a flat table and pull it back using a string with adult supervision.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 20: Show what you know

Observe the four teaching steps

1. BEFORE: A toy car is still on a table.

2. PUSH: A gentle hand sends it away.

3. PULL: A string draws it back.

4. COMPARE: Push points away; pull points toward.

Guided practice — answer in your exercise book

1. What force moves a trolley away from you?
2. What force brings a drawer toward you?
3. What is a force?
4. What can a push do to a light toy?
5. What can a pull do to a rope?
6. Which is a push action?
7. Which is a pull action?
8. Which practice is safe?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 21: Simple Machines Help Us

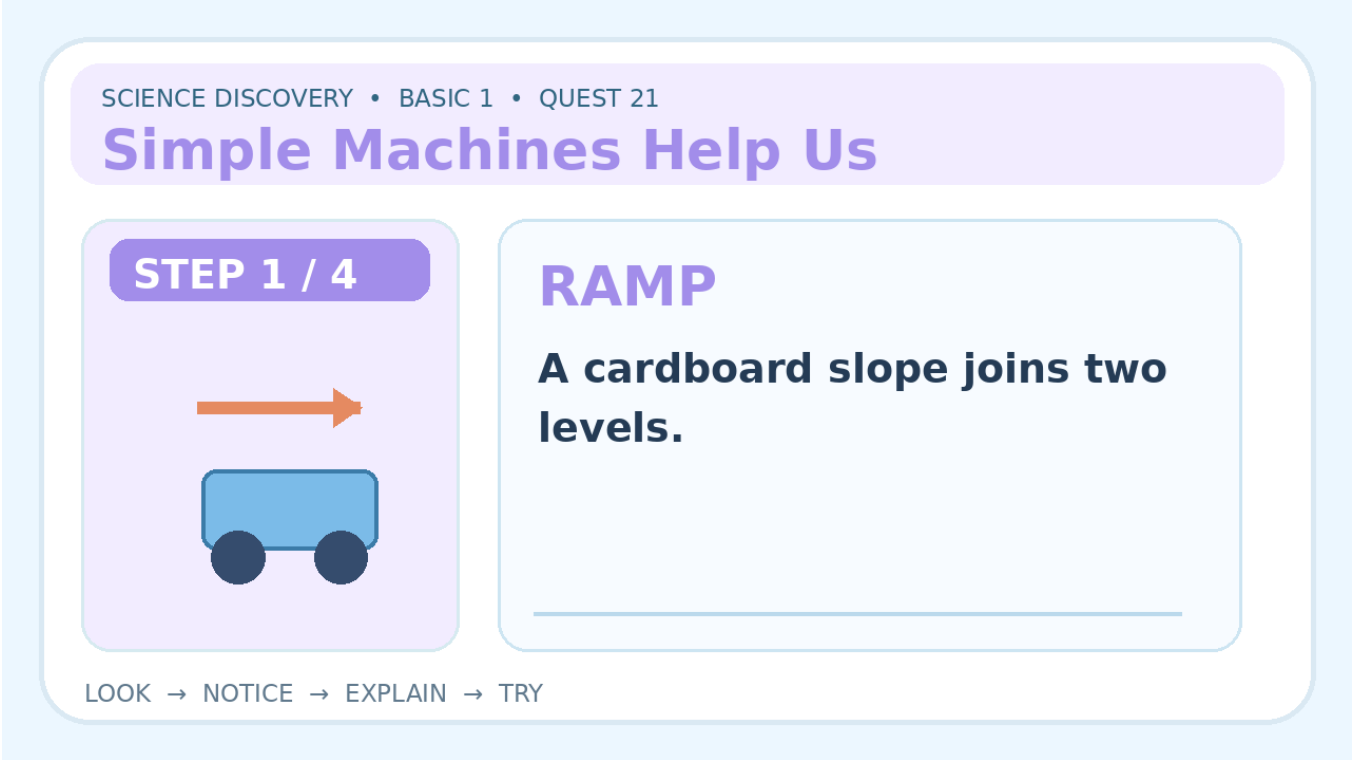
Forces and Energy | Recognise familiar simple machines and their uses.

LET’S LEARN

A simple machine is a tool that helps us do work. A ramp is an inclined plane, a seesaw uses a lever and a wheel can help objects roll. Simple machines make some jobs easier by changing how we apply a force. We should not play with sharp tools.

Science words to remember

Ramps • Levers • Wheels



Try it for yourself

Roll a small toy up a cardboard ramp with a teacher. Compare with lifting it straight up.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 21: Show what you know

Observe the four teaching steps

1. RAMP: A cardboard slope joins two levels.

2. ROLL: A toy moves up the slope.

3. LEVER: A seesaw tips around its middle support.

4. WHEEL: A cart rolls more easily using wheels.

Guided practice — answer in your exercise book

1. What can help a toy move up to a higher place?
2. Which playground object acts like a lever?
3. Which part helps a bicycle roll?
4. What is the use of a simple machine?
5. Which is an inclined plane?
6. What can help a cart move?
7. Which action is safest for a child learning about ramps?
8. Which example is a lever?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 22: Bathing to Stay Clean

Humans and the Environment | Explain why and how we bathe safely.

LET’S LEARN

Bathing washes sweat, dirt and germs from our skin. We use clean water, suitable soap and a clean towel. We should wash carefully, rinse soap away and dry ourselves. Never use very hot bathwater. Ask an adult for help with temperature and safety.

Science words to remember

Why bathe • Bathing steps • Bath safety



Try it for yourself

Arrange picture cards in order: prepare, wash, rinse and dry. Practise with a doll, not by getting into water at school.

Safety note: Use comfortably warm, not hot, water and appropriate adult supervision.

QUEST 22: Show what you know

Observe the four teaching steps

1. PREPARE: Clean water, soap and towel are ready.

2. WASH: A learner gently soaps the skin.

3. RINSE: Clean water washes away the soap.

4. DRY: A clean towel dries the body.

Guided practice — answer in your exercise book

1. Why do we bathe?
2. What should we use to wash skin?
3. What follows washing with soap?
4. What helps dry clean skin?
5. What should a child avoid?
6. What should be done before bathing?
7. What should follow rinsing?
8. Why is keeping skin clean helpful?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 23: Keep Teeth Clean

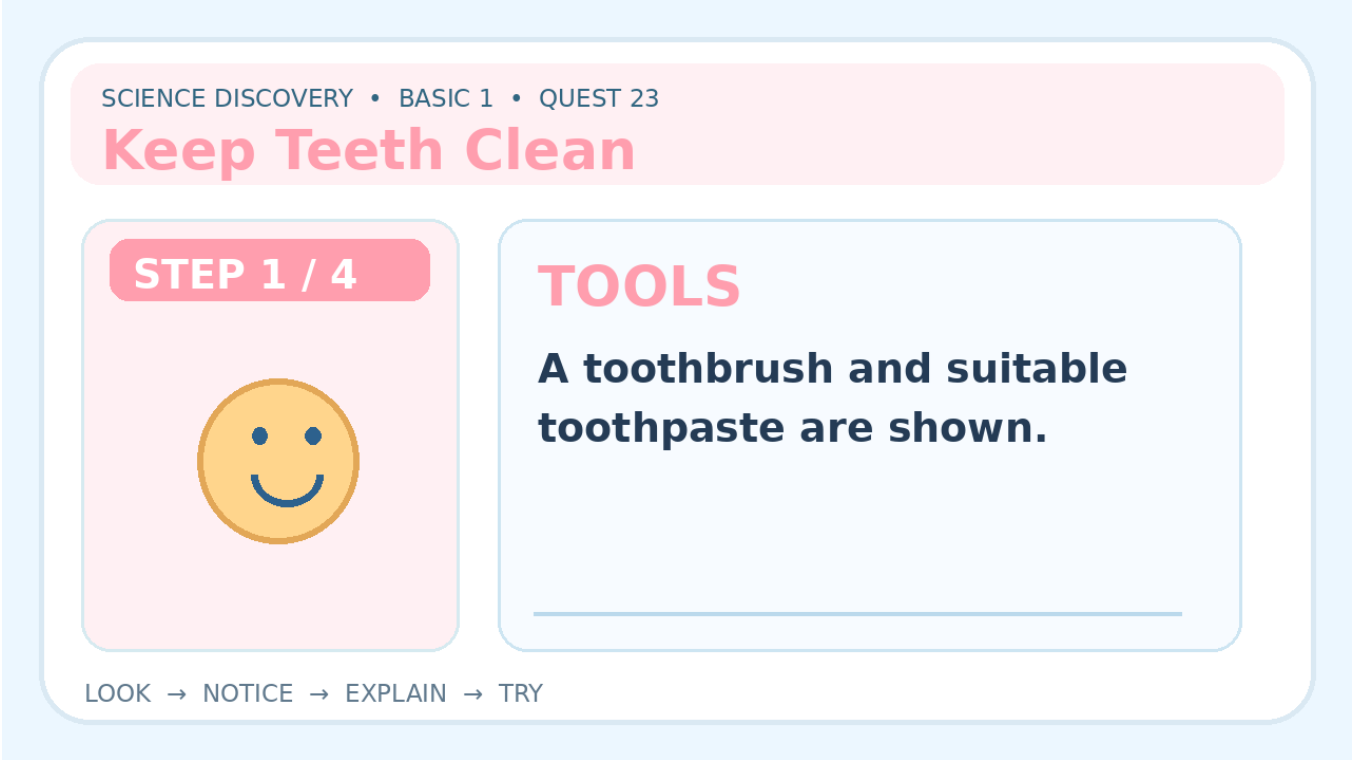
Humans and the Environment | Describe good toothbrushing habits.

LET’S LEARN

Brushing helps remove food particles and plaque from teeth. We use a toothbrush and fluoride toothpaste appropriate for a child. Brush gently twice each day, especially before sleeping. A trusted adult can help young children use the right amount of paste.

Science words to remember

Why brush • Materials • Routine



Try it for yourself

With a toothbrush model, show gentle brushing of front and back teeth. Ask an adult to check your routine at home.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 23: Show what you know

Observe the four teaching steps

1. TOOLS: A toothbrush and suitable toothpaste are shown.

2. BRUSH: Soft bristles move gently over teeth.

3. ROUTINE: Morning and bedtime symbols appear.

4. CHECK: A trusted adult helps with healthy brushing.

Guided practice — answer in your exercise book

1. What helps clean teeth?
2. What goes on a toothbrush?
3. How often should we normally brush?
4. What can brushing remove?
5. How should teeth be brushed?
6. When is an important time to brush?
7. Who may help a young child brush well?
8. What helps keep teeth healthy?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 24: Wash Hands the Right Way

Humans and the Environment | Demonstrate a safe and effective handwashing routine.

LET’S LEARN

Washing hands with soap and clean running water helps remove dirt and germs. Wet hands, rub with soap over palms and between fingers for about 20 seconds, rinse and dry. Wash before eating and after using the toilet. If clean running water is limited, ask an adult for a safe handwashing setup.

Science words to remember

Handwashing steps • When to wash • Hygiene



Try it for yourself

Demonstrate handwashing at a safe sink with teacher supervision. Sing a short song while rubbing with soap.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 24: Show what you know

Observe the four teaching steps

1. WET: Hands are under safe clean running water.

2. SOAP: Rub palms and between fingers for about 20 seconds.

3. RINSE: Clean water takes soap away.

4. DRY: Hands are dried before food is handled.

Guided practice — answer in your exercise book

1. What should you put on wet hands for washing?
2. What should you clean between?
3. What should happen after rubbing with soap?
4. When should we wash hands?
5. When else should we wash?
6. About how long should we rub with soap?
7. What helps dry clean hands?
8. Why wash hands with soap?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 25: Keep Our Environment Clean

Humans and the Environment | Explain why we keep homes, school and community clean.

LET’S LEARN

A clean environment is healthier and more pleasant. We can pick up litter safely, use bins, sweep floors and keep drains free of rubbish with adult help. We should never handle sharp or dangerous waste. Working together makes our school cleaner.

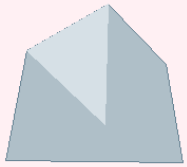
Science words to remember

Identify rubbish • Clean-up actions • Community care

SCIENCE DISCOVERY • BASIC 1 • QUEST 25

Keep Our Environment Clean

STEP 1 / 4



LOOK

Paper litter is scattered in a schoolyard.

LOOK → NOTICE → EXPLAIN → TRY

Try it for yourself

With a teacher, collect only safe paper litter using appropriate tools and put it in a bin. Draw before-and-after scenes.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 25: Show what you know

Observe the four teaching steps

1. LOOK: Paper litter is scattered in a schoolyard.

2. ACT: Learners put safe litter into bins.

3. AFTER: The schoolyard is clear and tidy.

4. CARE: Everyone follows safe waste rules.

Guided practice — answer in your exercise book

1. Where should paper litter go?
2. Which helps keep a classroom tidy?
3. What should you do about sharp waste?
4. What is a benefit of a clean school?
5. Which action protects drains?
6. What can a class do together?
7. Where should waste not be dumped?
8. How can you help at home?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 26: Care for Healthy Skin

Humans and the Environment | Recognise common skin problems and healthy hygiene habits.

LET’S LEARN

Skin can develop a rash, irritation or infection. Ringworm is a fungal infection, even though its name sounds like a worm. Eczema is not contagious. Many skin conditions look similar, and only a health professional can tell what is wrong. Keep skin clean, avoid sharing personal towels and tell an adult about worrying changes.

Science words to remember

Skin problems • Causes differ • Seeking help



Try it for yourself

Look at teacher-approved drawings of skin care choices. Sort good hygiene habits from unsafe habits; do not diagnose classmates.

Safety note: Do not diagnose, treat, scratch or photograph classmates’ skin conditions. Tell a trusted adult.

QUEST 26: Show what you know

Observe the four teaching steps

1. NOTICE: A child tells an adult about an itchy patch.

2. CARE: The child uses a clean personal towel.

3. LEARN: Ringworm is fungal; eczema is not contagious.

4. HELP: A health professional checks an unclear rash.

Guided practice — answer in your exercise book

1. Which is a common skin problem?
2. What type of infection is ringworm?
3. Is eczema something you catch from sitting beside someone?
4. Who should help when a rash is worrying or spreading?
5. Which habit may help protect skin?
6. What should a child avoid doing?
7. Why is medical advice sometimes needed?
8. What is kind behaviour toward someone with a rash?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 27: Technology Helps Communities

Humans and the Environment | Identify everyday technology and describe what it helps us do.

LET’S LEARN

Technology is the use of ideas, skills and tools to solve problems. A phone helps communication. A vehicle helps travel. A thermometer helps measure temperature. A farmer may use a watering can or other tool to grow crops. Use devices carefully and think about who benefits.

Science words to remember

Tools and uses • Society • Inventing safely



Try it for yourself

Design a simple tool using clean cardboard, bottle tops and tape with an adult. Explain the problem it helps solve.

Safety note: Work with a teacher or trusted adult where needed.

QUEST 27: Show what you know

Observe the four teaching steps

1. PROBLEM: Two people need to share a message.

2. TECHNOLOGY: A phone helps them talk.

3. MEASURE: A thermometer shows temperature.

4. INVENT: A safe cardboard model solves a small problem.

Guided practice — answer in your exercise book

1. What can a telephone help us do?
2. What helps people travel farther?
3. What does a thermometer help measure?
4. Why do people create tools?
5. What tool can a farmer use to carry water?
6. Which is a safe model-making material?
7. Who can benefit from helpful technology?
8. Which is an example of communication technology?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 28: Keeping Food for Later

Humans and the Environment | Identify common ways people process and preserve foods at home.

LET’S LEARN

Food can be changed by cooking, drying or other processing. Some foods are preserved by drying, smoking or refrigeration. Dried fish and dried pepper can last longer when stored properly. Different foods need different safe storage; food may still spoil. Never handle fire, hot oil or smoke during a class activity.

Science words to remember

Food processing • Preservation • Food safety



Try it for yourself

Compare pictures of fresh pepper and dried pepper. Ask an adult how food is safely kept at home.

Safety note: Food heating, frying, roasting and smoking must be demonstrated or handled by responsible adults.

QUEST 28: Show what you know

Observe the four teaching steps

1. FRESH: A tray of fresh peppers is shown.

2. DRY: Peppers are spread out for safe drying by adults.

3. STORE: Dried food is placed in a suitable dry container.

4. COMPARE: Refrigeration, drying and smoking are different methods.

Guided practice — answer in your exercise book

1. What can help keep some fish for longer?
2. Which appliance can keep suitable foods cold?
3. Which is an example of processed food?
4. Which food can be dried?
5. What is a safe role for a child near hot oil?
6. What may happen when food is not stored safely?
7. Which method uses smoke as part of preparation?
8. Who should handle ovens and fire during food preparation?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 29: Weather Watchers

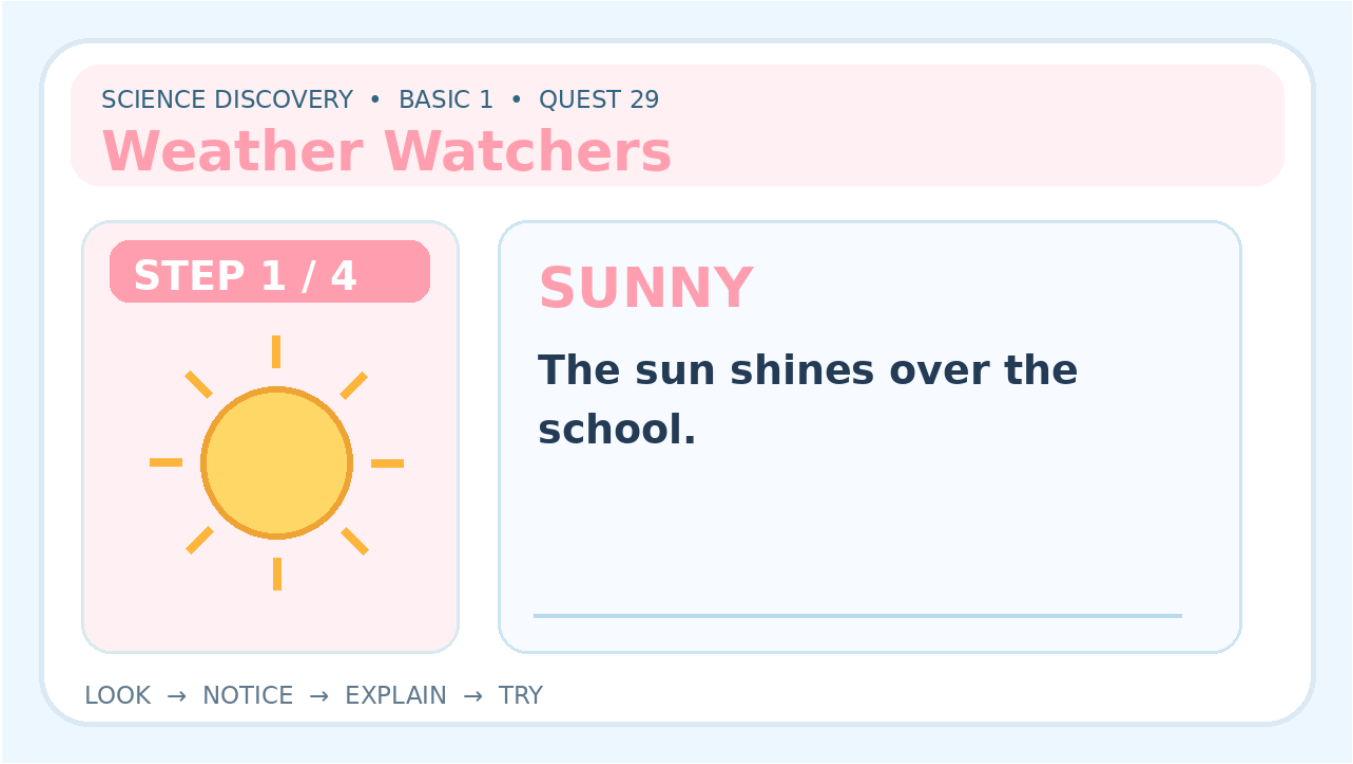
Humans and the Environment | Describe sunny, rainy, windy and cloudy weather.

LET’S LEARN

Weather is what the air and sky are like at a particular time. It may be sunny, cloudy, windy or rainy. Sunny days can feel warm; rainy days bring water from clouds. We choose suitable clothes and activities for each kind of weather. Do not go outside during dangerous lightning, strong wind or flooding.

Science words to remember

Weather conditions • Signs • Safe choices



Try it for yourself

Observe the sky safely from a doorway or covered place. Keep a weather chart for five days, using sun, cloud, wind or rain symbols.

Safety note: Observe storms from safe shelter. Do not go outside during lightning or flooding.

QUEST 29: Show what you know

Observe the four teaching steps

1. SUNNY: The sun shines over the school.

2. CLOUDY: Clouds cover much of the sky.

3. RAINY: Rain falls; people use raincoats.

4. WINDY: A flag and leaves move in the wind.

Guided practice — answer in your exercise book

1. Which weather has strong sunshine?
2. Which weather has water falling from clouds?
3. Which weather moves flags and leaves?
4. What covers much of the sky on a cloudy day?
5. Which item may help keep you dry in rain?
6. Which activity is unsafe during lightning?
7. What can children record on a weather chart?
8. How often can the weather change?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

Teacher and parent answer key

Quest 1: Look Closely at Our Surroundings

1. Eyes • 2. Its colour • 3. A plant • 4. A stone • 5. A goat • 6. A pencil • 7. Look closely at it • 8. Watch without touching

Quest 2: Plants and Animals in Ghana

1. Maize • 2. Butterfly • 3. Mango • 4. Chicken • 5. Cassava • 6. Goat • 7. Bird • 8. Leaf

Quest 3: What Living Things Need

1. Water • 2. Air • 3. Grass • 4. Sunlight • 5. Clean water • 6. Suitable water • 7. Shelter • 8. Air

Quest 4: Living, Non-living and Once-living

1. A growing seedling • 2. Plastic cup • 3. Dried leaf • 4. It cannot grow or feed itself • 5. A chick • 6. A hen • 7. A stone • 8. A dog

Quest 5: Materials Around Us

1. Wood • 2. Metal • 3. Glass • 4. Plastic • 5. Fabric • 6. Metal • 7. Ask an adult to help • 8. A pot

Quest 6: Sorting Materials by Their Features

1. Smooth and rough • 2. By colour • 3. Cotton wool • 4. Sandpaper • 5. Size • 6. Shape • 7. Metal • 8. We changed the sorting rule

Quest 7: Solids, Liquids and Gases

1. Solid • 2. Liquid • 3. Gas • 4. Water • 5. A pencil • 6. Air • 7. Milk • 8. Do not taste it

Quest 8: Mixing and Separating

1. Rice and beans together • 2. Two • 3. A sieve • 4. Paper shapes • 5. Hand-picking carefully • 6. Red and blue paper pieces • 7. You make a mixture • 8. One single spoon

Quest 9: Day and Night Repeat

1. Day • 2. Night • 3. Day • 4. Going to school • 5. Sun • 6. Night • 7. Repeatedly • 8. Day, night, day, night

Quest 10: The Sun Gives Us Light

1. Sun • 2. Torch • 3. Staring at the Sun • 4. Electric lamp • 5. Reflects it • 6. Daytime • 7. Sun • 8. More light reaches it

Quest 11: Where Did the Puddle Go?

1. A puddle • 2. It becomes smaller • 3. Liquid water changing to vapour • 4. Warmth • 5. Into the soil • 6. Flowing downhill • 7. Observe it • 8. Watch without drinking it

Quest 12: Water Sources and Uses

1. Borehole • 2. Washing hands • 3. Rain • 4. Close the tap • 5. Water that is safe to drink • 6. A well • 7. Cooking rice • 8. Ask an adult about safety

Quest 13: Air Is All Around Us

1. Air • 2. Wave a paper fan • 3. Flag • 4. Air • 5. Air takes up space inside • 6. Wave a cardboard fan • 7. No • 8. A gust of wind

Quest 14: My Body Parts

1. Eyes • 2. Ears • 3. Nose • 4. Hands • 5. Legs • 6. On the head • 7. Hand • 8. With respect

Quest 15: Homes of Living Things

1. Water • 2. Land • 3. Tree • 4. A living thing's home • 5. Food and shelter • 6. Near or in water • 7. Leave it undisturbed • 8. Soil

Quest 16: Energy Helps Us Do Things

1. Food • 2. Energy from batteries • 3. Energy • 4. Switch off unused lights • 5. To supply energy • 6. Energy from food • 7. Make the blades turn • 8. Clapping hands

Quest 17: Hot, Warm and Cold

1. Cold • 2. Very hot • 3. Warm • 4. Do not touch it • 5. In a refrigerator • 6. Make it warmer • 7. Ask an adult • 8. Cold

Quest 18: Electricity Around the Home

1. Electric fan • 2. Broom • 3. Refrigerator • 4. Television • 5. Keep away and tell an adult • 6. A metal object • 7. Electric lamp • 8. It helps suitable appliances work

Quest 19: Electronic Devices and Their Jobs

1. News and music • 2. Communicate • 3. Arithmetic • 4. Temperature • 5. Radio • 6. A connected electronic device • 7. Tablet • 8. Phone

Quest 20: Pushes and Pulls

1. Push • 2. Pull • 3. A push or a pull • 4. Move it • 5. Bring it toward you • 6. Closing a door by pressing it • 7. Opening a drawer toward you • 8. Move a light toy

Quest 21: Simple Machines Help Us

1. Ramp • 2. Seesaw • 3. Wheel • 4. To make some work easier • 5. Ramp • 6. Wheels • 7. Use cardboard and a toy • 8. Seesaw

Quest 22: Bathing to Stay Clean

1. To wash away dirt and sweat • 2. Clean water and suitable soap • 3. Rinsing • 4. Clean towel • 5. Very hot bathwater • 6. Check water is safe and comfortable • 7. Dry with clean towel • 8. It supports hygiene

Quest 23: Keep Teeth Clean

1. Toothbrush • 2. Suitable toothpaste • 3. Twice a day • 4. Food particles and plaque • 5. Gently • 6. Before bed • 7. Trusted adult • 8. Regular gentle brushing

Quest 24: Wash Hands the Right Way

1. Soap • 2. Fingers • 3. Rinse with clean water • 4. Before eating • 5. After using the toilet • 6. 20 seconds • 7. Clean towel or air drying • 8. To remove dirt and germs

Quest 25: Keep Our Environment Clean

1. Bin • 2. Sweeping • 3. Tell an adult • 4. Healthier surroundings • 5. Keep rubbish out • 6. Organise a safe clean-up • 7. In a water drain • 8. Put litter in a bin

Quest 26: Care for Healthy Skin

1. Rash • 2. Fungal • 3. No • 4. Trusted adult and health professional • 5. Use a clean personal towel • 6. Diagnosing a classmate • 7. Different problems can look similar • 8. Treat them respectfully

Quest 27: Technology Helps Communities

1. Communicate • 2. Vehicle • 3. Temperature • 4. To solve problems • 5. Watering can • 6. Cardboard • 7. People in a community • 8. Phone

Quest 28: Keeping Food for Later

1. Drying • 2. Refrigerator • 3. Roasted plantain • 4. Pepper • 5. Keep away and ask an adult • 6. It can spoil • 7. Smoking • 8. Responsible adult

Quest 29: Weather Watchers

1. Sunny • 2. Rainy • 3. Windy • 4. Clouds • 5. Raincoat • 6. Playing outside in an open field • 7. Today's weather • 8. From day to day

Source and rights note

Original independently authored study material aligned to indicators from NaCCA Science Basic 1–3 (2019). It is not a NaCCA publication or government endorsement. Scientific explanations, practice prompts, illustrations, and video teaching steps are originally produced for this package.